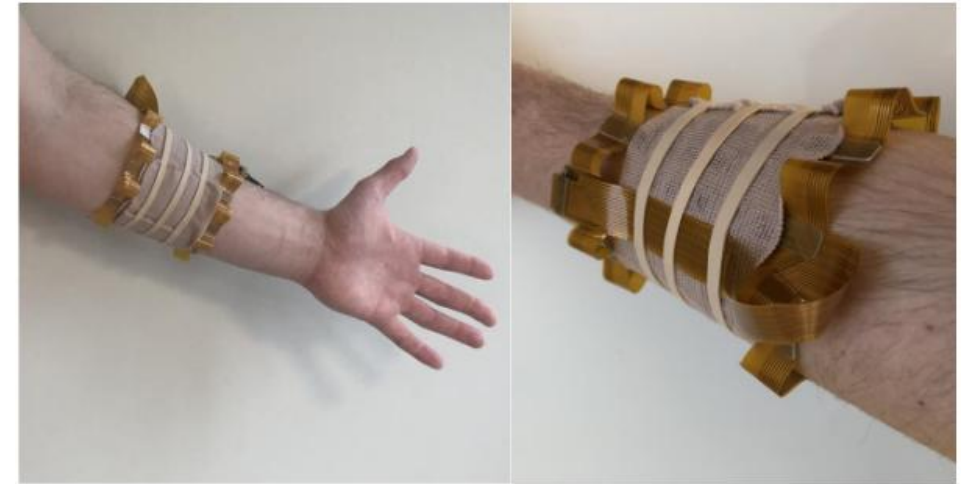


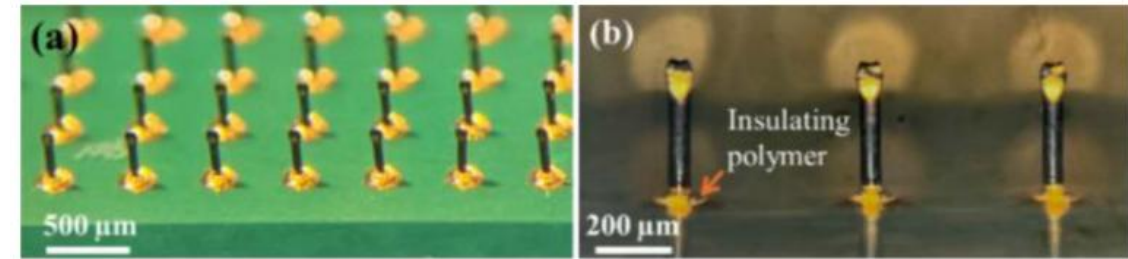
- Context: Control of hand prostheses through the analysis of myoelectric signals (generated by muscles)
- Use of electrodes for signal acquisition
- Common on the market:
 - Low number of electrodes for prosthesis control
- Problems
 - Low electrode count/density



Steeper Group, "Electrodes & Cables," *Steeper Group*. [Online]. [Accessed: Jul. 10, 2026]

- Project: Development of an ultra-high-density electrode array

- Up to 512 electrodes
- 5 mm between each electrode
- Microneedles for each electrode



- Objective:

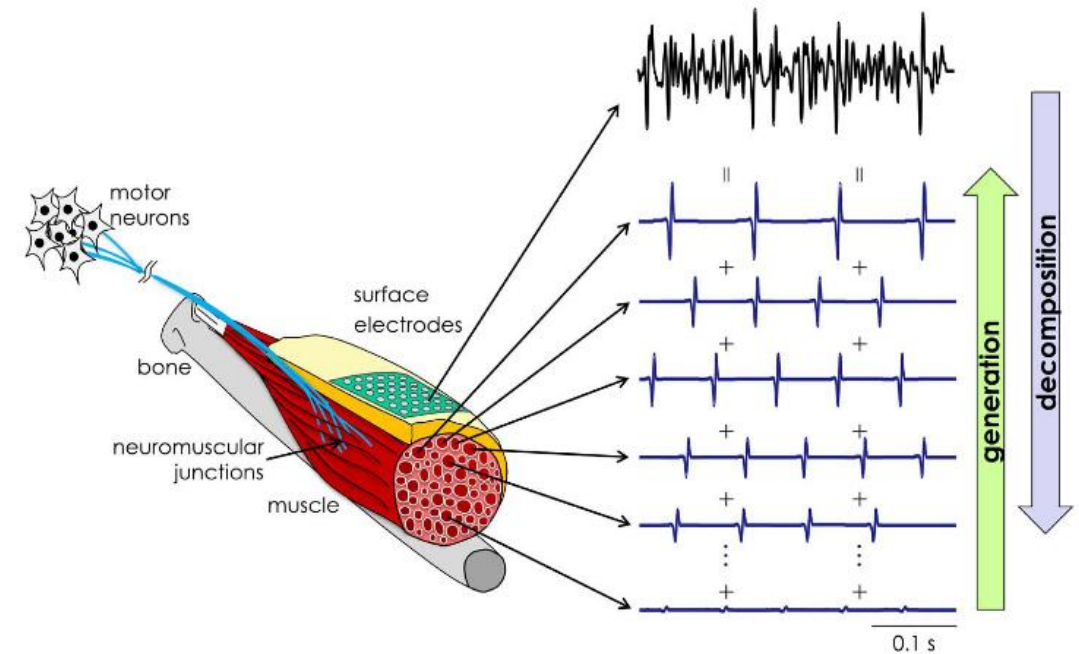
- Validate the performance of myoelectric signal analysis algorithms
- Test a myoelectric signal analysis method

- Decomposition of signals into motor neuron impulses (Motor Unit Action Potentials)

- Allows the behavior of motor neurons to be analyzed at the source

- Potential applications:

- Prostheses
- Muscle rehabilitation
- Diagnosis of neurological diseases



K. Haider, T. Tilly, V. Coyle, et C. Dalton, « Scalable and Customizable Trimaterial Microneedle Electrochemical Biosensor Platform With Cleanroom-Free Fabrication », *IEEE Sens. Lett.*, vol. 9, n° 9, p. 1-4, sept. 2025

D. Farina et A. Holobar, « Characterization of Human Motor Units From Surface EMG Decomposition », *Proc. IEEE*, vol. 104, n° 2, p. 353-373, févr. 2016